www.RocketRevise.com

Questions

Q1.

In cells, energy can be released from substrates by anaerobic and aerobic respiration.

(a) Most eukaryotic cells can respire anaerobically.

(i) Which of the following describes a step in the process of anaerobic respiration?

(1)

RocketRevise

☒ **A** decarboxylation of lactate

☒ **B** phosphorylation of hexoses

☒ **C** oxidation of pyruvate

☒ **D** removal of phosphate groups from glucose

(ii) What happens to the lactate concentration during a period of anaerobic respiration?

(1)

☒ **A** decreases, causing a decrease in blood pH

☒ **B** decreases, causing an increase in blood pH

☒ **C** increases, causing a decrease in blood pH

☒ **D** increases, causing an increase in blood pH

(b) Most eukaryotic cells are also able to respire aerobically.

(i) How do respiratory substrates enter the Krebs cycle?

RocketRevise

(1)

☒ **A** as molecules containing 2 carbon atoms produced by the link reaction

☒ **B** as molecules containing 3 carbon atoms produced by the link reaction

☒ **C** as molecules containing 2 carbon atoms produced by RUBISCO

☒ **D** as molecules containing 3 carbon atoms produced by RUBISCO

(ii) Draw a diagram of a mitochondrion.

Label only the part of the mitochondrion where the Krebs cycle occurs.

(2)

www.RocketRevise.com

www.RocketRevise.com

(iii) Describe the role of chemiosmosis in the synthesis of ATP.

(5)

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 10 marks)

RocketRevise

Q2.

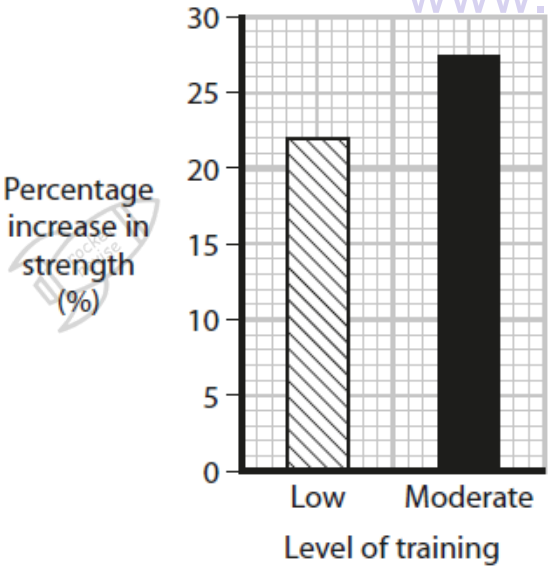
Many athletes train to improve muscle strength.

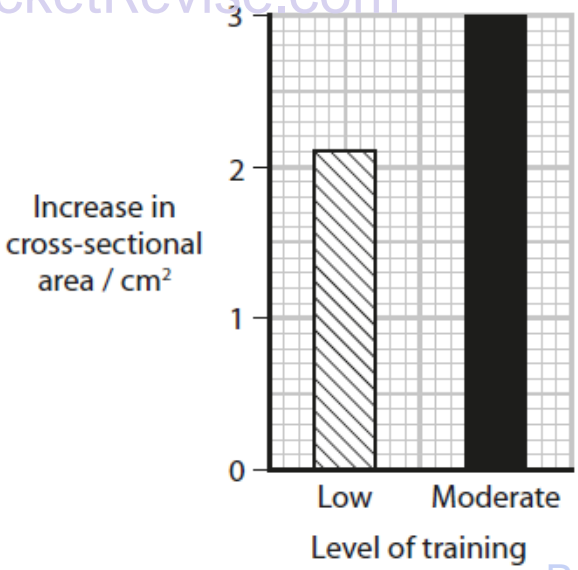
In an experiment, the effect on a muscle of 12 weeks of low or moderate training was investigated.

The graphs show the effect of 12 weeks of training on muscle strength and muscle cross-sectional area.

www.RocketRevise.com

www.RocketRevise.com





RocketRevise

(a) The muscle strength of the athlete before moderate training was 12.5 a.u.

Calculate the muscle strength of the athlete after 12 weeks of moderate training.

Give your answer to 3 significant figures.

(3)

Answer .....

(b) The types of muscle fibres present in samples taken before the training and after 12 weeks of low and moderate training were determined.

Fibre type	Percentage of fibres (%)		
	Before training	After 12 weeks of training	
		Low	Moderate
Slow twitch	44	44	44
Fast twitch	6	2	1
Intermediate	50	54	55

RocketRevise

Intermediate fibres have structural properties between those of slow and fast twitch fibres.

(i) Describe how the structure of an intermediate fibre will differ from the structure of a slow twitch fibre.

(3)

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

(ii) Comment on the effect of exercise on this muscle.

Use the information in the graphs and the table to support your answer.

(3)

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 9 marks)

RocketRevise

Q3.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Aerobic respiration involves four main stages:

glycolysis, link reaction, Krebs cycle and oxidative phosphorylation.

(a) One product of all three stages shown in the table is ATP (adenosine triphosphate).

Complete the table to show one other product of each stage.

www.RocketRevise.com

www.RocketRevise.com

(3)

Stage	One other product
glycolysis	
Krebs cycle	
oxidative phosphorylation	

RocketRevise

(b) An incomplete equation for the aerobic respiration of a substrate is shown below.

Complete the equation by inserting the numbers on the dotted lines to balance it.

(2)

$$C_{57}H_{104}O_6 + \ldots\ldots O_2 \rightarrow \ldots\ldots CO_2 + 52\ H_2O$$

(c) One molecule of glucose contains 2867.48 kJ of energy.

One molecule of glucose generates a maximum of 38 molecules of ATP in aerobic respiration.

A molecule of ATP contains 30.51 kJ of usable energy. Usable energy is available for chemical reactions in the cell.

(i) Which reaction releases energy from ATP?

(1)

☒ **A** the conversion of ADP to ATP

☒ **B** the phosphorylation of ADP

☒ **C** the hydrolysis of ATP

RocketRevise

☒ **D** the removal of adenosine molecules from ATP

(ii) Calculate the maximum percentage of energy in a glucose molecule that can be converted to usable energy in ATP.

(2)

Answer .....

(iii) Describe what happens to the energy that is not converted to usable energy in a muscle cell.

(2)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

(d) Describe how oxygen is involved in the production of ATP on the cristae in the mitochondria.

(2)

RocketRevise

.....

.....

.....

.....

.....

.....

(Total for question = 12 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Animals use aerobic and anaerobic respiration to produce ATP.

(a) Aerobic respiration has three stages where reactions occur between enzymes and substrates.

(i) Which row of the table shows the location of each stage?

(1)

www.RocketRevise.com

www.RocketRevise.com

	Glycolysis	Link reaction	Krebs cycle
☒ A	cytoplasm	cytoplasm	cytoplasm
☒ B	cytoplasm	mitochondrial matrix	mitochondrial matrix
☒ C	mitochondrial matrix	cytoplasm	cytoplasm
☒ D	mitochondrial matrix	mitochondrial matrix	mitochondrial matrix

(ii) A student calculated the volume of oxygen used by a cockroach as 1.7 cm^3 $hour^{-1}$. The same cockroach produced 1.3 cm^3 $hour^{-1}$ of carbon dioxide in the same time.

What is the respiratory quotient (RQ) of this cockroach?

(1)

RocketRevise

☒ **A** 0.7

☒ **B** 0.8

☒ **C** 1.3

☒ **D** 2.2

(iii) Hydramethylnon is an insecticide that is used to treat termite infestations.

Hydramethylnon inhibits the electron transport chain.

Which row of the table shows the expected changes caused by hydramethylnon?

(1)

	Oxygen consumption	Lactate production
☒ A	decrease	decrease
☒ B	decrease	increase
☒ C	increase	decrease
☒ D	increase	increase

RocketRevise

(b) The table shows the oxygen consumption and body mass of two animals.

Animal	Oxygen consumption / $cm^3 h^{-1}$	Body mass / g
Southern elephant seal	1.82×10^4	5.00×10^6
Field mouse	20.00	10.00

These figures can be used to calculate the metabolic rate of each animal in cm^3 g^{-1} h^{-1}.

Calculate how many times greater the metabolic rate of the field mouse is compared with that of the southern elephant seal.

www.RocketRevise.com

www.RocketRevise.com

(2)

Answer .....

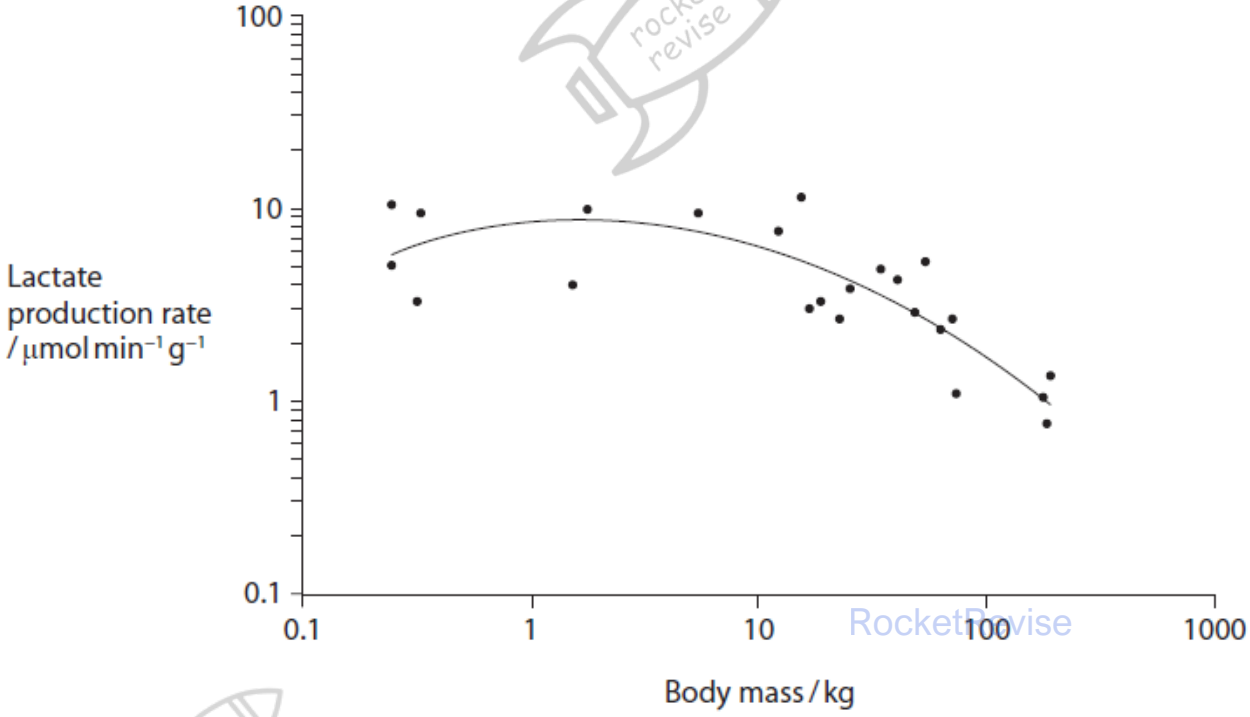
* (c) The photograph shows a crocodile (*Crocodylus porosus*) that lives in saltwater habitats.



(Source: © Danny Ye/Shutterstock)

An investigation measured the lactate production rate of these crocodiles when they were active.

The graph shows the results of this investigation.



Each plotted point on the graph represents a single crocodile.

Discuss the roles of the different types of respiration in these active crocodiles.

Use the information in the graph and your own knowledge to support your answer.

(6)

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 11 marks)

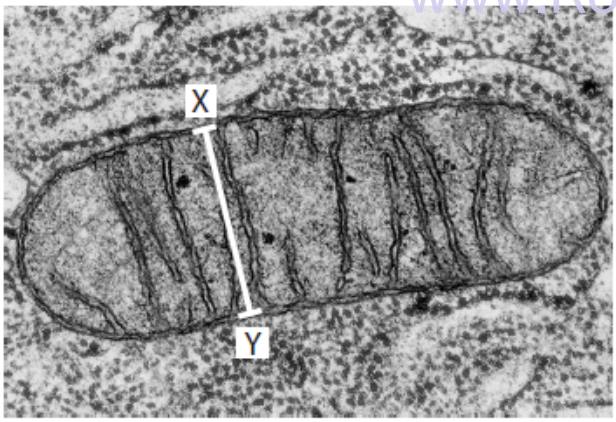
RocketRevise

Q5.

The photograph shows a transmission electron micrograph of a mitochondrion and surrounding cytoplasm.

www.RocketRevise.com

www.RocketRevise.com



(Source: © CNRI / SCIENCE PHOTO LIBRARY)

(a) The mitochondrion shown in the diagram is 2000 nm in length.

Calculate the width of this mitochondrion between X and Y.

RocketRevise

(2)

Answer ..... nm

(b) Describe how ATP is synthesised by oxidative phosphorylation in the mitochondrion.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

(c) Describe how ATP is used to supply energy for biological processes.

(2)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

(Total for question = 7 marks)

Q6.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Mitochondria supply energy for the processes in cells.

(a) (i) How many of these statements about the inner membrane of a mitochondrion are correct?

(1)

- it is folded (has cristae)
- it is the location of ATP synthase
- it is the site of the electron transfer chain
- it is a site of protein synthesis

☒ **A** 1

☒ **B** 2

☒ **C** 3

☒ **D** 4

(ii) Anaerobic respiration produces lactate as a waste product.

Which is the result of lactate production?

(1)

☒ **A** blood becomes less acidic

☒ **B** muscle becomes less fatigued

☒ **C** pH of the blood increases

☒ **D** muscle becomes more acidic

www.RocketRevise.com

www.RocketRevise.com

(b) Describe how chemiosmosis is involved in the synthesis of ATP.

(2)

.....

.....

.....

.....

.....

.....

RocketRevise

(c) A culture containing respiring cells was supplied with radioactive oxygen molecules.

Which will be the first substance to contain radioactive oxygen atoms?

(1)

☒ **A** ATP

☒ **B** carbon dioxide

☒ **C** reduced NAD

☒ **D** water

(Total for question = 5 marks)

Q7.

RocketRevise

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Many of the metabolic processes taking place in cells require ATP.

(a) The electron transport chain (ETC) is involved in the production of ATP.

(i) Where is the site of the electron transport chain?

(1)

☒ **A** cytoplasm

☒

www.RocketRevise.com

www.RocketRevise.com

B inner mitochondrial membrane

☒ **C** mitochondrial matrix

☒ **D** outer mitochondrial membrane

(ii) Enzymes are involved in the production of ATP by the electron transport chain.

Which type of molecule is an enzyme?

(1)

☒ **A** carbohydrate

☒ **B** lipid

RocketRevise

☒ **C** phospholipid

☒ **D** protein

(iii) Describe the role of reduced NAD in aerobic respiration.

(2)

.....

.....

.....

.....

.....

.....

(b) Carbon monoxide and cyanide affect the production of ATP.

(i) Carbon monoxide binds to the haem group of haemoglobin.

RocketRevise

Explain how carbon monoxide will affect the production of ATP.

(3)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

(ii) Cyanide stops the transfer of electrons along the electron transport chain.

Explain why cyanide does not affect the production of ATP from anaerobic respiration.

(2)

RocketRevise

.....

.....

.....

.....

.....

.....

(Total for question = 9 marks)

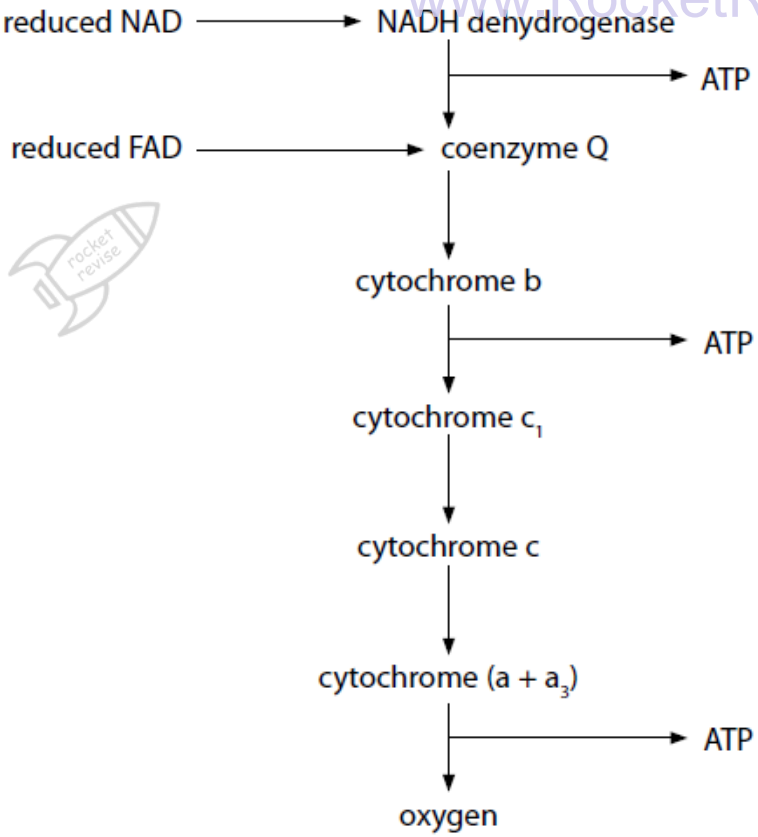
Q8.

The diagram shows the sequence of electron carriers in the electron transport chain.
The diagram also shows the stages where ATP is generated.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



(a) Where is the site of the electron transport chain?

(1)

☒ **A** cytoplasm

☒ **B** inner mitochondrial membrane

☒ **C** matrix

☒ **D** outer mitochondrial membrane

(b) The table gives some information about inhibitors of the electron transport chain.

Inhibitor	Site of inhibition
Cyanide	between cytochrome (a + a_3) and oxygen
Rotenone	between NADH dehydrogenase and coenzyme Q

(i) Explain the effect that cyanide will have on ATP production by the electron transport chain.

(2)

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

(ii) Metabolism of succinate results in the production of reduced FAD.

Explain the effect that rotenone will have on ATP production by the electron transport chain.

RocketRevise

(2)

.....

.....

.....

.....

.....

(iii) Explain why these inhibitors will not affect the production of ATP from anaerobic respiration. Use the information in the table to support your answer.

(2)

.....

.....

.....

.....

RocketRevise

.....

(Total for question = 7 marks)

Q9.

The scientific document you have studied is adapted from the article '*Brain aging and neurodegeneration: from a mitochondrial point of view*' by Amandine Grimm and Anne Eckert.

www.RocketRevise.com

www.RocketRevise.com

Use the information from the article and your own knowledge to answer the following questions.

(a) Suggest why there is a relationship between the ageing process and neurodegeneration (paragraphs 1 and 2).

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(b) Describe how energy released from glucose is used in oxidative phosphorylation (paragraph 2).

(3)

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(c) Explain how free radicals (reactive oxygen species, ROS) can cause damage to mitochondrial DNA
(paragraphs 2 and 4).

www.RocketRevise.com

www.RocketRevise.com

(3)

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

(d) "The brain is a remarkable organ composed by highly differentiated cells... with different morphology, according to their role and their localization" (paragraph 6).

Describe how this differentiation can occur.

(3)

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

(e) Explain what is meant by the phrase 'mitochondrial proteome' (paragraph 7).

(2)

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

(f) Explain why synaptic mitochondria are more susceptible to ageing than the mitochondria found in other parts of a neurone (paragraph 9).

(2)

.....

.....

.....

.....

.....

.....

(g) Explain why the mitochondrial genome is essential for oxidative phosphorylation (paragraph 10).

(2)

.....

.....

.....

.....

.....

.....

(h) Suggest how damaged mitochondria are removed by 'autophagy' (paragraph 11).

(2)

.....

.....

.....

.....

.....

.....

(Total for question = 20 marks)

www.RocketRevise.com

www.RocketRevise.com

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Aerobic respiration is made up of several stages.

(a) Which is the correct order of the stages of respiration required for the complete oxidation of glucose?

RocketRevise

(1)

☒	**A**	glycolysis	Krebs cycle	link reaction
☒	**B**	glycolysis	link reaction	Krebs cycle
☒	**C**	link reaction	Krebs cycle	glycolysis
☒	**D**	Krebs cycle	glycolysis	link reaction

(b) Write a **balanced** equation for the complete oxidation of glucose in aerobic respiration.

(2)

.....

(c) The equation shows the aerobic respiration of a lipid.

$$2\ C_{18}H_{34}O_2 + 51\ O_2 \rightarrow 36\ CO_2 + 34\ H_2O$$

(i) Which is the respiratory quotient when this lipid is used in aerobic respiration?

(1)

☒ **A** 0.67

☒ **B** 0.71

☒ **C** 1.06

☒ **D** 1.42

(ii) The complete oxidation of one molecule of this lipid produces 122 molecules of ATP.

One molecule of this lipid contains 12 244 kJ of energy.

www.RocketRevise.com

www.RocketRevise.com

One molecule of ATP yields 30.51 kJ of the energy.

Calculate the percentage of energy in this lipid molecule that can be converted to energy in ATP.

Give your answer to **three** significant figures.

(2)

Answer ..... %

(Total for question = 6 marks)

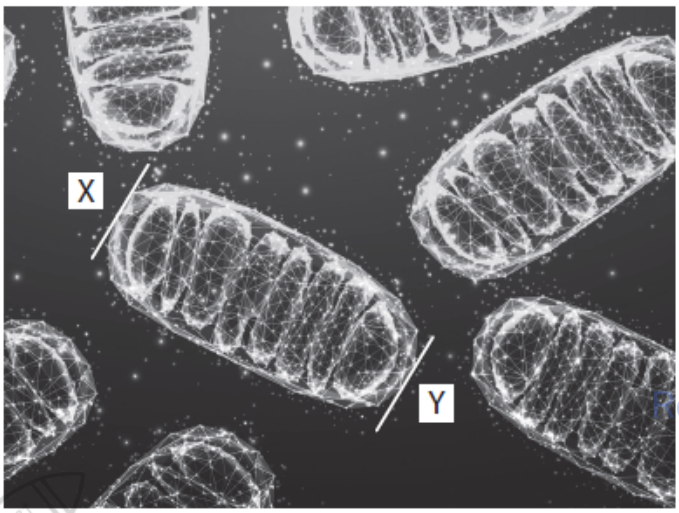
RocketRevise

Q11.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some stages of aerobic respiration take place inside mitochondria.

The photograph shows some mitochondria.



(Source: © Inkoly / Shutterstock)

(a) (i) Name a structure formed by the folding of the inner mitochondrial membrane.

(1)

.....

(ii) Which is a role of the inner mitochondrial membrane?

(1)

www.RocketRevise.com

www.RocketRevise.com

☒ **A** to convert glucose to pyruvate

☒ **B** to phosphorylate ADP

☒ **C** to produce glucose from carbon dioxide

☒ **D** to trap light

(iii) The actual length of the mitochondrion between X and Y in the photograph is 2.7 µm.

Calculate the magnification of this mitochondrion.

Give your answer in standard form.

(2)

RocketRevise

Answer .....

(b) Describe how **reduced** NAD is involved in the synthesis of ATP inside a mitochondrion.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) (i) Which is the role of oxygen molecules in mitochondria?

RocketRevise

(1)

☒ **A** to accept electrons

☒ **B** to donate electrons

☒ **C** to donate hydrogen ions

☒ **D** to form carbon dioxide

(ii) Which part of a mitochondrion contains the electron transport chain (ETC)?

(1)

☒

www.RocketRevise.com

www.RocketRevise.com

A inner mitochondrial membrane

☒ **B** mitochondrial matrix

☒ **C** outer mitochondrial membrane

☒ **D** ribosome

(Total for question = 9 marks)

RocketRevise

RocketRevise

www.RocketRevise.com